

Splunk 101 Capstone

SOC Investigation Report - Ryan Adams Incident

Analyst	Nguyen Thai Hoang
Primary host	FRONTDESK-PC1.KCD.local
Primary user	Ryan.Adams
Investigation window	2025-10-15 12:45:00 - 13:15:00 UTC
Data sources	Windows Security, Sysmon, Zeek, Suricata, Windows Application

Findings

- **Primary account:** Ryan.Adams.
- **Authentication pattern:** 62 failed Event ID 4625 records for ryan.adams from 172.16.0.184 / DESKTOP-924H12, followed by Event ID 4624 Type 3 network logon activity.
- **Suspicious file:** C:\Users\Ryan.Adams\Music\python.exe was created at 12:57:00 UTC and executed at 13:00:33 UTC.
- **External communication:** python.exe connected from 172.16.0.110 to 157.245.46.190:8888. Zeek later corroborated an established TLSv1.3 session to the same destination.
- **ADDC01 activity:** python.exe resolved ADDC01.KCD.local to 172.16.0.7 and connected to TCP/135 and TCP/49669. This is RPC-related; successful lateral movement is not proven.
- **Persistence:** PowerShell launched schtasks.exe to create PythonUpdate, configured to run python.exe at system startup as SYSTEM.
- **Defender correlation:** Windows Application logs later reported SECURITY_PRODUCT_STATE_SNOOZED; causation is not established.

Investigation Summary

Within the 12:45-13:15 UTC window, Windows Security logs showed concentrated failed logons for Ryan.Adams from 172.16.0.184 / DESKTOP-924H12, followed by Type 3 network logon activity. Sysmon then recorded creation of C:\Users\Ryan.Adams\Music\python.exe at 12:57:00 UTC and execution at 13:00:33 UTC with explorer.exe as parent.

Immediately after execution, Sysmon attributed several actions to python.exe: an external connection to 157.245.46.190:8888, a DNS lookup for ADDC01.KCD.local resolving to 172.16.0.7, and RPC-related connections to 172.16.0.7 on TCP/135 and TCP/49669. Zeek corroborated an established TLSv1.3 session to 157.245.46.190:8888.

At 13:04:59 UTC, PowerShell launched schtasks.exe to create the PythonUpdate startup task running python.exe as SYSTEM. This strongly supports persistence. The evidence supports host compromise and suspicious external communication, but does not prove RDP, encoded PowerShell, successful lateral movement, a specific malware family, or sustained C2 beaconing.

Who / What / When / Where / Why / How

WHO - Ryan.Adams on FRONTDESK-PC1.KCD.local; source authentication workstation 172.16.0.184 / DESKTOP-924H12; external IP 157.245.46.190; internal host ADDC01.KCD.local / 172.16.0.7.

WHAT - Suspicious authentication activity followed by python.exe creation/execution, external communication, DNS/RPC activity involving the domain controller, and scheduled-task persistence.

WHEN - 2025-10-15, approximately 12:52:08-13:05:53 UTC. Primary search window: 12:45-13:15 UTC.

WHERE - FRONTDESK-PC1.KCD.local; suspicious file path C:\Users\Ryan.Adams\Music\python.exe.

WHY - The exact motive and method by which credentials became usable are unknown. The behavior is consistent with unauthorized access, external control/communication, domain-oriented interaction, and persistence.

HOW - Failed/successful network logons were followed by creation and execution of a suspicious executable, process-attributed network/DNS activity, and creation of a SYSTEM startup scheduled task.

Key Timeline

UTC	Activity	Evidence / Interpretation
12:52:08	Failed logons	62 Event ID 4625 records for ryan.adams from 172.16.0.184 / DESKTOP-924H12.
12:52:12	Network logon	Event ID 4624 Type 3 records for Ryan.Adams from the same source.
12:57:00	python.exe created	C:\Users\Ryan.Adams\Music\python.exe; prior investigation linked creation to chrome.exe.
13:00:33	python.exe executed	Sysmon Event ID 1; parent explorer.exe.
13:00:34	External connection	python.exe -> 157.245.46.190:8888.
13:00:34	DNS lookup	python.exe queried ADDC01.KCD.local -> 172.16.0.7.
13:00:34-35	RPC-related	python.exe -> 172.16.0.7:135 and :49669.
13:03:29	TLS corroboration	Zeek observed established TLSv1.3 to 157.245.46.190:8888.
13:04:59	Persistence	PowerShell -> schtasks.exe /create /tn PythonUpdate /tr python.exe /sc onstart /ru SYSTEM /f.
13:05:53	Defender state	SECURITY_PRODUCT_STATE_SNOOZED; causation not proven.

Recommendations

- **Contain:** isolate FRONTDESK-PC1.KCD.local and revoke active Ryan.Adams sessions.
- **Credentials:** reset Ryan.Adams credentials, review other accounts targeted from 172.16.0.184 / DESKTOP-924H12, and enforce MFA where supported.
- **Eradicate:** preserve evidence, remove the PythonUpdate task, and quarantine/remove python.exe after collection.
- **Review:** examine PowerShell, scheduled-task, browser, Defender, and endpoint artifacts for additional payloads or persistence.
- **Network:** block or closely monitor 157.245.46.190 and search historical telemetry for the same IP/port.
- **Domain controller:** review ADDC01 logs for authentication, RPC, remote execution, service/task creation, and account changes to confirm or refute lateral movement.
- **Detection engineering:** alert on executables from user media folders, SYSTEM startup tasks, high-volume failed logons followed by success, and unusual external connections from nonstandard binaries.

Assessment Boundaries / Unknowns

- The successful authentication records shown are Logon Type 3 (network), not Logon Type 10 (RDP); therefore this report does not claim brute-force RDP.
- The observed PowerShell persistence command is not encoded.
- The connection to 157.245.46.190:8888 is suspicious and process-attributed, but sustained C2 beaconing was not established.
- RPC-related connections to ADDC01 do not by themselves prove successful lateral movement or domain-controller compromise.
- The Defender SNOOZED event is temporally correlated but causation is not demonstrated.

This report is based on simulated capstone telemetry available in Splunk and demonstrates SOC investigation, correlation, scoping, and evidence-based reporting.